

Absolutely. I went through the **actual PPTX you uploaded**, so I'll make the script follow your **exact slide order and content**, rather than the earlier proposed structure. Your PPT has 20 slides, with the first half covering background through methodology and the second half covering results, web application, conclusion, and references.

I suggest:

- **Member 1: Slides 1–11**
- **Member 2: Slides 12–20**

This gives a natural handover at **Proposed Methodology**, which is a good point to switch speakers.

The language below is deliberately **very simple and easy to memorize**, but still professional.

□ **PRESENTATION SCRIPT**

□ **MEMBER 1 — Slides 1–11**

Slide 1 — Title

Good morning everyone.

We are here to present our research project titled:

“Pediatric vs Adult Dengue Severity Prediction with Subgroup-Specific SHAP Explanations.”

I am [Name], and this is my teammate [Name].

Our work focuses on using **machine learning and explainable AI** to study dengue severity differently for children and adults.

Our supervisor is **Mr. Mohammad Arif Hasan Chowdhury**, Assistant Professor, Department of CSE, USTC.

So, let's begin.

Slide 2 — Overview

First, I will briefly explain the background of our work, the problem we found, our objectives, and some previous research.

Then, my teammate will explain our methodology, results, web application, and conclusion.

Slide 3 — Background

Dengue severity models are usually trained using patients of all ages together.

But children and adults are not exactly the same.

Their bodies can respond differently to dengue.

So, when we put everyone into one model, some important differences between children and adults may be hidden.

This is the main idea behind our research.

We wanted to see whether building separate models for children and adults could give us better and more useful information.

Slide 4 — Introduction

Dengue is a disease spread mainly by **Aedes mosquitoes**.

It is a major health problem in many tropical and subtropical countries.

One important challenge is that some patients can become severely ill.

So, finding high-risk patients early is very important.

In our research, we focus on one important question:

Can machine learning understand dengue severity differently for children and adults?

Slide 5 — Introduction

Most previous machine-learning studies use **age as just another input feature**.

In other words, they give age to the model, but they still train **one model for everyone**.

But in real clinical situations, doctors may consider children and adults differently.

For example, children may experience rapid plasma leakage, while adults may have more comorbidities and different bleeding patterns.

So, in our thesis, we compare:

Pooled models

with

Pediatric and adult models.

We also compare their **SHAP explanations**.

Slide 6 — Problem Statement

Now we come to the main problem.

We identified several issues.

First, many models use **pooled data** for children and adults.

Second, SHAP explanations are often also calculated for the whole dataset.

This can hide differences between age groups.

Third, a single model may give one general decision boundary, even though children and adults may have different clinical patterns.

And finally, many research models are not connected to a useful system where we can enter patient information and see an explanation.

So our research question is:

Do separate pediatric and adult models give different predictions and explanations compared with one pooled model?

And can we make this useful through a web dashboard?

Slide 7 — Objectives

Our work has three main goals.

First, we developed pediatric, adult, and pooled machine-learning models for dengue severity prediction.

Second, we generated **SHAP explanations** for the different groups and identified important model-derived feature splits.

Third, we created an **explainable web dashboard** where the models and explanations can be used and patient data can be stored.

Slide 8 — Related Works

Before developing our model, we studied previous research.

Earlier studies used machine learning to predict dengue diagnosis and severity.

For example, some researchers used **decision trees**, while others focused specifically on **pediatric patients**.

Some studies also found that age can be an important factor.

However, these studies did not directly compare **pediatric and adult machine-learning models with subgroup-specific explanations**.

Slide 9 — Related Works

More recent studies started using more advanced machine-learning methods.

They used models such as:

Random Forest, ANN, GBM, SVM, and ensemble models.

Some of them also used **SHAP** to explain their models.

But most of these studies still used a **pooled population**.

So, their SHAP explanation was mainly a **global explanation**.

This means they explained the whole dataset instead of separately explaining children and adults.

Slide 10 — Related Works

Other studies also used SHAP and ensemble learning.

These studies achieved good prediction results.

However, we noticed one common issue:

Most of them did not study age-conditioned SHAP explanations.

So, there was still a question:

Are the important features and their effects really the same for children and adults?

This is where our research focuses.

Slide 11 — Proposed Methodology

Now I will briefly introduce our overall methodology.

We started with the dengue dataset.

Then we performed **preprocessing and feature engineering**.

After that, we divided the data into:

Pediatric, Adult, and Pooled cohorts.

We trained and evaluated machine-learning models for each cohort.

Then we used **SHAP** to understand the model decisions.

Finally, we connected the trained models and explanations to our **early-stage dengue risk assessment dashboard**.

Handover

That covers the background, problem, objectives, related works, and overall methodology of our research.

Now I would like to hand over the presentation to my teammate, who will explain the dataset, model performance, SHAP results, web application, and conclusion.

□ MEMBER 2 — Slides 12–20

Slide 12 — Methodology: Dataset & Split

Thank you.

Now I will continue with our methodology.

We started with **1,329 raw dengue records**.

After filtering based on the **NS1 test**, we obtained **972 NS1-positive records**.

We then divided them into three groups.

The **pediatric group** contained **158 patients aged 18 or below**.

The **adult group** contained **814 patients above 18 years**.

We also kept a **pooled group containing all 972 patients**.

For each group, we used an **80:20 stratified train-test split**.

Slide 13 — Model Performance

Now let's look at our model results.

We tested different machine-learning models and compared their performance.

For the **pediatric group**, the best accuracy was **90.6%**.

For the **adult group**, Random Forest achieved **84.7% accuracy**.

For the **pooled group**, Random Forest achieved **79.0% accuracy**.

So, we can see that the age-stratified models performed better than the pooled model in our dataset.

This is an important finding from our research.

Slide 14 — SHAP Feature Attribution

Next, we looked at **which features were important to the models**.

For this, we used **SHAP**, which helps us understand why a machine-learning model makes a prediction.

One important finding was that **lymphocyte percentage** was very important in both the pediatric and adult models.

This tells us that the models were not only making predictions.

They were also showing us which clinical features were influencing those predictions.

Slide 15 — Model-Derived SHAP Feature Splits

This is one of the interesting parts of our research.

We extracted feature splits from the SHAP results.

For the pediatric model, the lymphocyte split was:

greater than 44.50%.

For the adult model, it was:

greater than 24.99%.

We also found other differences.

For example, the pediatric model had splits for **neutrophils and WBC**, while the adult model had splits for **HCT and age**.

This suggests that the models may use clinical features differently for different age groups.

However, these are model-derived values, not clinical treatment thresholds.

Slide 16 — Web Application

Now we move to our web application.

We developed an **early-stage dengue risk assessment dashboard**.

The main purpose is to make our machine-learning model easier to use.

A user can enter patient information.

The system can then use the appropriate model and provide a risk prediction.

The system also provides an explanation using **SHAP**.

So, instead of only saying:

“This patient is high risk,”

the system can also show:

“These features contributed to the prediction.”

Slide 17 — Web Application

Our dashboard also allows us to keep patient assessment records.

This is useful because a patient can be assessed more than once.

The system can store these different assessments.

So, in the future, these stored records can help us build a proper **longitudinal or time-based prediction model**.

At the current stage, however, the stored observations are not yet used as sequential model inputs.

Slide 18 — Conclusion & Future Work

Now let's summarize our work.

We found that **age-stratified models performed better than the pooled model** in our dataset.

The pediatric model achieved **90.6% accuracy**, while the adult model achieved **84.7%**.

We also found a strong similarity between pediatric and adult SHAP feature rankings, with a **Kendall's tau of 0.900**.

But even though the rankings were similar, the model-derived feature splits were different.

This suggests that a pooled model may hide some **age-specific decision behaviour**.

Future work

There are four main things we want to do next.

First, test the models on **larger and independent datasets**.

Second, collect a **larger pediatric dataset**.

Third, use **WHO-adjudicated severity labels** instead of our current platelet-derived proxy.

And finally, use the stored data to develop **true longitudinal models** in the future.

Slide 19 — References

These are some of the important papers and sources that we used in our research.

They include previous dengue machine-learning studies, SHAP research, and studies on pediatric and adult dengue.

We used these works to understand the existing research and identify the gap that our work addresses.

Slide 20 — Final / Thank You

That brings us to the end of our presentation.

Our main message is:

Children and adults may not behave the same way in dengue prediction, so studying them separately can reveal useful information that a pooled model may hide.

Thank you for listening.

We are happy to answer any questions.

☐ **Very simple speaking strategy**

Don't try to memorize every word exactly.

For each slide, remember **3 things**:

Member 1

Slides 1–5:

- ☐ *What is dengue + why age matters*

Slides 6–7:

- ☐ *What is the problem + what are our goals*

Slides 8–10:

- ☐ *What others did + what they missed*

Slide 11:

- ☐ *What we did*

Member 2

Slide 12:

- ☐ *What data we used*

Slides 13–15:

- ☐ *What the models found*

Slides 16–17:

- ☐ *What our web app does*

Slide 18:

- ☐ *What we learned + what's next*

Slides 19–20:

- ☐ *References + thank you*

★ Three sentences to remember if you get nervous

If either of you forgets something during the presentation, come back to these three sentences:

“We wanted to know whether children and adults behave differently in dengue severity prediction.”

“So, we trained separate models and used SHAP to understand their decisions.”

“Our results show that age-specific models can reveal differences that may be hidden in a pooled model.”

Those three sentences basically tell the **whole story of your thesis**.